

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{2}{3} g_3^4 \text{LF}_{3,0} [m_3] \delta_{i1i2} \delta_{i3i4} + g_3^4 \text{LF}_{4,-1} [m_3] \delta_{i1i2} \delta_{i3i4} - \frac{16}{15} g_3^4 \text{LF}_{5,-2} [m_3] \delta_{i1i2} \delta_{i3i4} + \right. \\ & \frac{2}{9} \sum_p g_3^4 \text{LF}_{3,0} [m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{12} \sum_p g_3^4 \text{LF}_{4,-1} [m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{45} \sum_p g_3^4 \text{LF}_{5,-2} [m_d^p] \delta_{i1i2} \delta_{i3i4} + \frac{4}{9} \sum_p g_3^4 \text{LF}_{3,0} [m_q^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{6} \sum_p g_3^4 \text{LF}_{4,-1} [m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{16}{45} \sum_p g_3^4 \text{LF}_{5,-2} [m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{9} \sum_p g_3^4 \text{LF}_{3,0} [m_u^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{12} \sum_p g_3^4 \text{LF}_{4,-1} [m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{8}{45} \sum_p g_3^4 \text{LF}_{5,-2} [m_u^p] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{54} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{216} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{216} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{216} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{216} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{8} g_2^2 g_3^2 \text{LF}_{2,1,0} [m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{8} g_2^2 g_3^2 \text{LF}_{2,2,-1} [m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{4} g_2^2 g_3^2 \text{LF}_{3,1,-1} [m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{8} g_2^2 g_3^2 \text{LF}_{4,1,-2} [m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{8} g_2^2 g_3^2 \text{LF}_{2,1,0} [m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{8} g_2^2 g_3^2 \text{LF}_{2,2,-1} [m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{4} g_2^2 g_3^2 \text{LF}_{3,1,-1} [m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{8} g_2^2 g_3^2 \text{LF}_{4,1,-2} [m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_3^4 \text{LF}_{2,1,0} [m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{36} g_3^4 \text{LF}_{2,2,-1} [m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{25}{36} g_3^4 \text{LF}_{3,1,-1} [m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{9} g_3^4 \text{LF}_{4,1,-2} [m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_3^4 \text{LF}_{2,1,0} [m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{36} g_3^4 \text{LF}_{2,2,-1} [m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{25}{36} g_3^4 \text{LF}_{3,1,-1} [m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{9} g_3^4 \text{LF}_{4,1,-2} [m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_3^4 \text{LF}_{2,1,0} [m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{36} g_3^4 \text{LF}_{2,2,-1} [m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{25}{36} g_3^4 \text{LF}_{3,1,-1} [m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{9} g_3^4 \text{LF}_{4,1,-2} [m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_3^4 \text{LF}_{2,1,0} [m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{36} g_3^4 \text{LF}_{2,2,-1} [m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{25}{36} g_3^4 \text{LF}_{3,1,-1} [m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{9} g_3^4 \text{LF}_{4,1,-2} [m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{3} g_3^2 \bar{y}_d^{i2p} y_d^{i1p} \text{LF}_{2,1,0} [m_d^p, \tilde{\mu}] \delta_{i3i4} + \\ & \frac{1}{6} g_3^2 \bar{y}_d^{i2p} y_d^{i1p} \text{LF}_{2,2,-1} [m_d^p, \tilde{\mu}] \delta_{i3i4} + \frac{1}{6} g_3^2 \bar{y}_d^{i2p} y_d^{i1p} \text{LF}_{3,1,-1} [m_d^p, \tilde{\mu}] \delta_{i3i4} - \\ & \frac{1}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_d^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_d^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} g_3^4 \text{LF}_{2,1,0} [m_d^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_3^4 \text{LF}_{3,1,-1} [m_d^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_d^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_d^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} g_3^4 \text{LF}_{2,1,0} [m_d^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_3^4 \text{LF}_{3,1,-1} [m_d^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \bar{y}_d^{i2i3} y_d^{i1i4} \bar{a}_e^{pr} a_e^{pr} \text{LF}_{3,1,0} [m_l^p, m_e^r] + 3 \bar{y}_d^{i2i3} y_d^{i1i4} \bar{a}_e^{pr} a_e^{pr} \text{LF}_{4,1,-1} [m_l^p, m_e^r] - \\ & 2 \bar{y}_d^{i2i3} y_d^{i1i4} \bar{a}_e^{pr} a_e^{pr} \text{LF}_{5,1,-2} [m_l^p, m_e^r] - 3 \bar{y}_d^{i2i3} y_d^{i1i4} \bar{a}_d^{pr} a_d^{pr} \text{LF}_{3,1,0} [m_q^p, m_d^r] + \\ & 9 \bar{y}_d^{i2i3} y_d^{i1i4} \bar{a}_d^{pr} a_d^{pr} \text{LF}_{4,1,-1} [m_q^p, m_d^r] - 6 \bar{y}_d^{i2i3} y_d^{i1i4} \bar{a}_d^{pr} a_d^{pr} \text{LF}_{5,1,-2} [m_q^p, m_d^r] - \\ & \frac{2}{3} g_3^2 \bar{y}_d^{pi3} y_d^{pi4} \text{LF}_{2,1,0} [m_q^p, \tilde{\mu}] \delta_{i1i2} + \frac{1}{3} g_3^2 \bar{y}_d^{pi3} y_d^{pi4} \text{LF}_{2,2,-1} [m_q^p, \tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{3} g_3^2 \bar{y}_d^{pi3} y_d^{pi4} \text{LF}_{3,1,-1} [m_q^p, \tilde{\mu}] \delta_{i1i2} - \frac{1}{108} g_1^2 g_3^$$